

Disaster Response Coordination – Group 5

Scope of the Societal Problem

We want to build a database for disaster relief so organizations and field workers can track operations, building damage, and ground reports in one place across affected areas. It should show who is running which operation, how bad the building and regional damage is, the details and reliability of field reports, and the current status of individuals (like missing or rescued) along with who reported them.

How does the Data Interacts

1. An organization can conduct many operations, but each operation is run by exactly one organization.
2. A location can host many operations, but each operation takes place at one location.
3. A location can contain many buildings, but each building stands in exactly one location.
4. A person can file many reports, but each report is filed by exactly one person.
5. A location can be the subject of many reports, but each report concerns exactly one location.

The entities

objects of the database: Organization, Location, Operations, Building, Person, Report.

The Relationships

1. Organization --conducts--> Operations
2. Location --hosts--> Operations
3. Location --contains--> Building
4. Person --files--> Report
5. Location --is subject of--> Report

Attributes for each entity

1. ORGANIZATION: org_name, org_type, contact_info — to know who is responding and how to reach them.
2. LOCATION: location_name, region, damage_level — damage_level lets coordinators prioritize which regions need resources most urgently.
3. OPERATIONS: operation_type, start_date, status — status (Active/Completed) shows which relief efforts are still ongoing.
4. BUILDING: building_type, damage_status — tracks structural damage per building, feeding into a location's overall damage_level.
5. PERSON: name, age, status — status (missing, rescued, injured) is the key field responders need to act on.
6. REPORT: report_text, timestamp, reliability_score — reliability_score lets coordinators weigh field reports when conflicting information comes in.

PKs and FKs

7. ORGANIZATION — PK: org_id
8. LOCATION — PK: location_id
9. OPERATIONS — PK: operation_id; FKs: org_id → ORGANIZATION, location_id → LOCATION
10. BUILDING — PK: building_id; FK: location_id → LOCATION
11. PERSON — PK: person_id
12. REPORT — PK: report_id; FKs: person_id → PERSON, location_id → LOCATION

Sample Data for the ERD

Before Normalization

Table 1 — Operations Log (unnormalized)

op_id	op_type	start	status	buildings_affected	org_id	org_name	org_type	contact	loc_id	location	region	damage
OP1	Search & Rescue	2026-01-05	Active	B12 Warehouse–Severe; B14 Residential–Moderate	ORG1	Rode Kruis	NGO	+31201234567	L1	Sector 4 Warehouse District	North Region	Severe
OP2	Medical Aid	2026-01-06	Active	B12 Warehouse–Severe; B14 Residential–Moderate	ORG1	Rode Kruis	NGO	+31201234567	L1	Sector 4 Warehouse District	North Region	Severe
OP3	Structural Assessment	2026-01-06	Completed	B09 Bridge Structure–Cracked	ORG2	Brandweer Rotterdam	Government	+31108765432	L2	Riverside Bridge	North Region	Moderate
OP4	Debris Removal	2026-01-07	Active	B12 Warehouse–Severe; B14 Residential–Moderate; B15 Residential–Minor	ORG1	Rode Kruis	NGO	+31201234567	L1	Sector 4 Warehouse District	North Region	Severe
OP5	Shelter Setup	2026-01-07	Active	B20 School–Minor	ORG3	GGD Amsterdam	Medical	+31205551212	L3	Central School Shelter	South Region	Minor
OP6	Water Distribution	2026-01-08	Completed	B09 Bridge Structure–Cracked	ORG2	Brandweer Rotterdam	Government	+31108765432	L2	Riverside Bridge	North Region	Moderate

Table 2 — Reports Log (unnormalized)

person_id	loc_id	name	age	status	location	region	damage	report_text	timestamp	reliability
P1	L1	Anna de Vries	34	Reported safe	Sector 4 Warehouse District	North Region	Severe	Roof collapsed, 3 trapped	2026-01-05 08:15	0.9
P1	L2	Anna de Vries	34	Reported safe	Riverside Bridge	North Region	Moderate	Bridge cracked, unsafe	2026-01-05 09:40	0.8
P2	L1	Tom Bakker	61	Field Volunteer	Sector 4 Warehouse District	North Region	Severe	Additional survivors found near warehouse	2026-01-06 13:20	0.75
P3	L3	Els Jansen	45	Reported safe	Central School Shelter	South Region	Minor	Shelter operational, 40 people housed	2026-01-07 15:00	0.95
P4	L1	Youssef El Amrani	29	Field Volunteer	Sector 4 Warehouse District	North Region	Severe	Debris cleared from main road	2026-01-07 09:30	0.8
P1	L3	Anna de Vries	34	Reported safe	Central School Shelter	South Region	Minor	Additional medical supplies needed	2026-01-08 10:10	0.7

After Normalization

ORGANIZATION

org_id	org_name	org_type	contact
ORG1	Rode Kruis	NGO	+31201234567
ORG2	Brandweer Rotterdam	Government	+31108765432
ORG3	GGD Amsterdam	Medical	+31205551212

LOCATION

loc_id	location	region	damage
L1	Sector 4 Warehouse District	North Region	Severe
L2	Riverside Bridge	North Region	Moderate
L3	Central School Shelter	South Region	Minor

OPERATIONS

op_id	op_type	start	status	org_id	loc_id
OP1	Search & Rescue	2026-01-05	Active	ORG1	L1
OP2	Medical Aid	2026-01-06	Active	ORG1	L1
OP3	Structural Assessment	2026-01-06	Completed	ORG2	L2
OP4	Debris Removal	2026-01-07	Active	ORG1	L1
OP5	Shelter Setup	2026-01-07	Active	ORG3	L3
OP6	Water Distribution	2026-01-08	Completed	ORG2	L2

BUILDING

building_id	building_type	damage_status	loc_id
B12	Warehouse	Severe	L1
B14	Residential	Moderate	L1
B15	Residential	Minor	L1
B09	Bridge Structure	Cracked	L2
B20	School	Minor	L3

PERSON

person_id	name	age	status
P1	Anna de Vries	34	Reported safe
P2	Tom Bakker	61	Field Volunteer
P3	Els Jansen	45	Reported safe
P4	Youssef El Amrani	29	Field Volunteer

REPORT

report_id	person_id	loc_id	report_text	timestamp	reliability
R1	P1	L1	Roof collapsed, 3 trapped	2026-01-05 08:15	0.9
R2	P1	L2	Bridge cracked, unsafe	2026-01-05 09:40	0.8
R3	P2	L1	Additional survivors found near warehouse	2026-01-06 13:20	0.75
R4	P3	L3	Shelter operational, 40 people housed	2026-01-07 15:00	0.95
R5	P4	L1	Debris cleared from main road	2026-01-07 09:30	0.8
R6	P1	L3	Additional medical supplies needed	2026-01-08 10:10	0.7

Week 5 – Normalization review

In week 5 we loaded two real datasets: the official casualty and damage figures of the 2015 Nepal earthquake (75 districts) and the FAO EVE flood monitoring data (bi-weekly flooded area and exposed population per district, 2024–2026). The sample data above only had the violations we added on purpose. The real files contained the following normal form violations.

Normal form	Violation	Where in the data	How it is fixed in our database
1NF	Two columns with the same name	Earthquake file: two columns are both called "Unknown" (unknown-sex deaths and unknown-sex injuries). In a relation every attribute name must be unique.	Renamed to deaths_unknown and injured_unknown in EARTHQUAKE_IMPACT.
1NF	A row that is not a district	Earthquake file: the last row holds national totals and has no DIST_ID or district name, so it has no key value and is a different kind of row than all others.	Not loaded. Only used to check that the loaded district rows add up to the same totals.
1NF	Cell with a stray value	Earthquake file: Displaced_Females for Humla holds only a line break; the column is empty in every other row.	Displaced columns not loaded.
2NF	Partial dependency	Flood file: a row is identified by (adm2_pcode, period_number). start_date and end_date depend on period_number alone; adm2_name, adm1_pcode and adm1_name depend on adm2_pcode alone.	Period dates in the new table FLOOD_PERIOD; district details in LOCATION; only the measured values stay in FLOOD_OBSERVATION.
2NF	Partial dependency in our own first draft	Our first FLOOD_OBSERVATION had key (location_id, period_start) and also stored period_end, which depends on period_start only.	period_end moved to FLOOD_PERIOD (key period_start).
3NF	Transitive dependency	Earthquake file: DIST_ID → ZONE → ZONE_CODE → REG_CODE. The zone decides the codes, not the district.	Zones and codes not stored; the region (province) is stored once per district in LOCATION.
3NF	Transitive dependency	Flood file: adm2_pcode → adm1_pcode → adm1_name (district → province code → province name).	Only the province name is stored, once per district (LOCATION.region).
3NF	Derived (calculated) columns	Earthquake file: Tot_Deaths and Total_Injured are sums of the female, male and unknown columns, and are wrong in 3 districts (e.g. Lalitpur, Nuwakot). Flood file: the *_sq_km columns are the hectare columns divided by 100, and perc_* columns are calculated from area.	Not stored. Totals come from the view V_EARTHQUAKE_IMPACT; square km and percentages can be calculated when needed.

Changes to the design

New tables: EARTHQUAKE_IMPACT (one row per district), FLOOD_PERIOD (one row per bi-weekly period) and FLOOD_OBSERVATION (one row per district per period with flooding), plus the view V_EARTHQUAKE_IMPACT for totals.

LOCATION: location_name is now UNIQUE, population and households were added (NULL when unknown), and region holds the province.

Result

After these changes the loaded tables are in 3NF: each table has one key and every other column depends only on that whole key. One redundancy remains by design: location.damage_level is derived from the earthquake figures when the data is loaded.